Operating instructions

V 1.0 English

QiTech Laser

QiTech Industries GmbH
Landwehrstraße 55
64293 Darmstadt

EC Declaration of Conformity

According to 2006/42/EC, Annex II, No. 1 A

The content of the EC Declaration of Conformity corresponds to Annex II, No. 1 A of Directive 2006/42/EC.

Manufacturer of the machine:

QiTech Industries GmbH
Landwehrstraße 55
64293 Darmstadt

Representative of the manufacturer:

Milan von dem Bussche
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64293 Darmstadt

Person authorised to compile the technical file:

Milan von dem Bussche
Landwehrstraße 55
64293 Darmstadt

General designation	Laser diameter measuring device
Type	Laser diameter measuring device
Function	Laser measurement of filament diameter
Model	QiTech Laser
Trade name	QiTech Laser
Serial number	

We hereby declare that the above-mentioned machine is in conformity with all relevant provisions of the EC Machinery Directive 2006/42/EC. The machine is also in compliance with all relevant provisions of the following EC Directives:

- 2014/30/EU Electromagnetic Compatibility / European EMC Directive
- 2014/35/EU Low Voltage Directive (protection objectives only)
- 2011/65/EU ROHS Directives (ROHS 2)

The following harmonized standards (or parts of these standards) have been applied:

- DIN EN ISO 12100
- DIN EN ISO 20607
- DIN EN 60825-1

Darmstadt, 08.09.2026



Milan von dem Bussche

Read before commissioning!

These operating instructions are intended to enable all people working on the machine to carry out their work safely and correctly. Every person who operates, maintains or repairs the machine must have read the operating instructions, particularly the safety regulations, before commissioning. These operating instructions must always be available on the machine. The operator must ensure that all information necessary for the safe operation of the machine is made available to the operating personnel. If the operating instructions are available in electronic form, the operator must ensure that the operating personnel are able to view the operating instructions. The original version of these operating instructions is written in English. We ask you to read through the relevant instructions carefully and to observe them thoroughly. After unpacking the machine, check for any damage that may have occurred during transport.

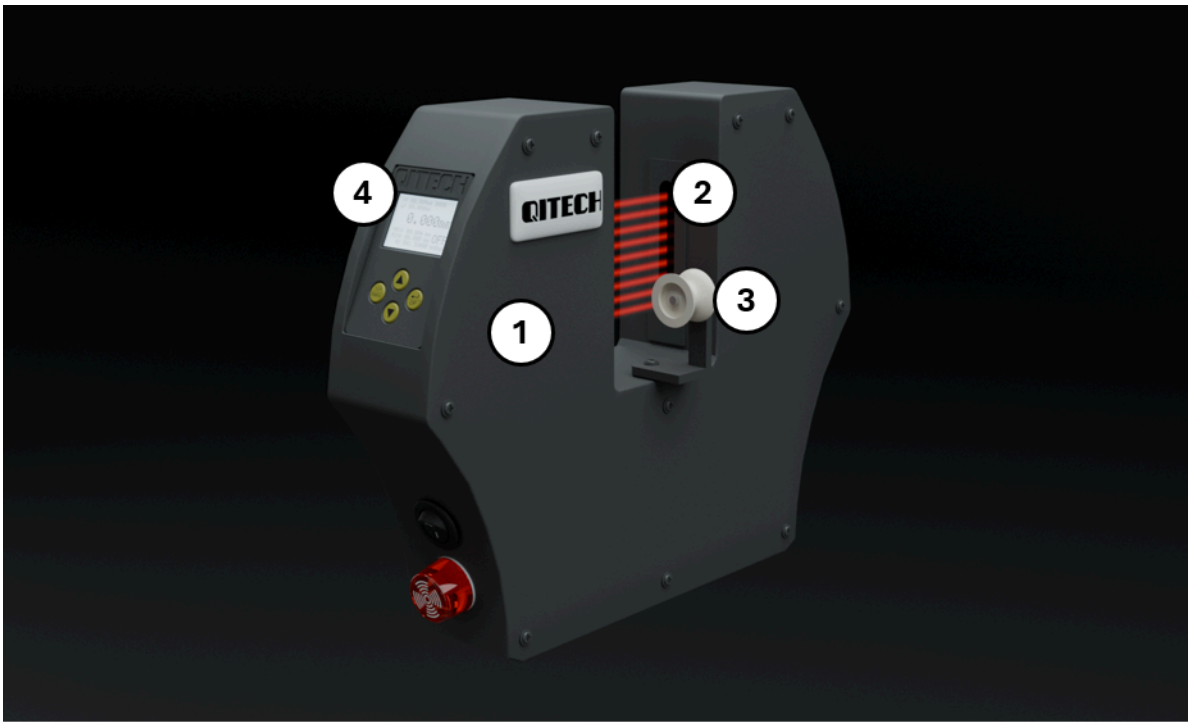
NOTICE: Before using the machine for the first time, please inform yourself about the national regulations (e.g. occupational safety, environmental restrictions). Reproduction and reprinting in any form, even in extracts, requires the written authorization of QiTech Industries GmbH.

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1. Machine description

1.1 Component overview







Item number	Description
1	Measuring head
2	Measuring gap

3	Guide rollers
4	Measured-value display
5	Stand column
6	Control display
7	On/off switch (rocker switch)
8	Acoustic-optical signal device (tolerance alarm)

The QiTech Laser is switched on and off by means of the on/off rocker switch on the measuring head. The device has no main switch with emergency switching-off function and no emergency stop, as it is a measuring device without hazardous movement.

The QiTech Laser is operated as part of a process line together with the QiTech Extruder, QiTech Airpath or QiTech Aquapath and the QiTech Winder. It is supplied with power via its own mains plug and is connected to the control display of the process line via a steel-sheathed Z-Link cable (8-pin), where the laser-specific view of the QiTech Control software is shown.

The arrangement of the QiTech Laser in the process line is shown in Annex A.

1.2 Technical specifications

Dimensions and weight	
Length x Width x Height	160 mm x 360 mm x min. 1050, max. 1330 mm (adjustable height)
Weight	14 kg
Electrical specifications	
Rated voltage	220 V AC
Rated frequency	50 Hz
Power consumption	< 50 W

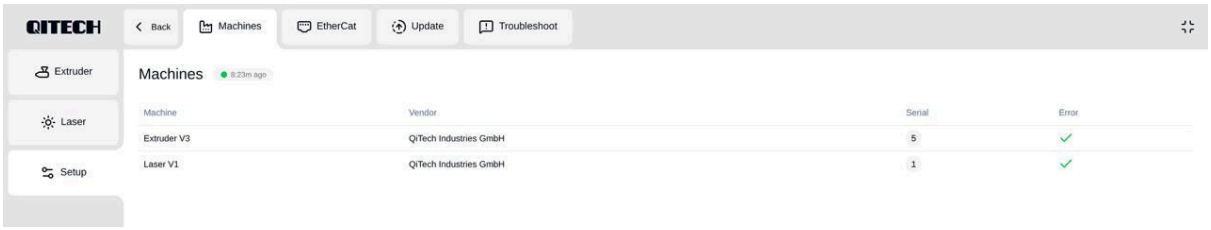
Rated apparent power	< 0,05 kVA
Rated current	< 0,23 A
Measuring parameters	
Measuring range (diameter)	0,2 – 30 mm
Measuring accuracy	± 2 µm
Resolution	1 µm
Measuring rate	> 50 m/s
Ambient temperature	4 - 40 °C
Ambient humidity	< 85% without condensed dew
Laser class (per DIN EN 60825-1)	Class 2
Maximum output power	< 50 W
Noise emissions	
Workplace-related emission value (L_pA)	< 50 dB

1.3 Controls and interface

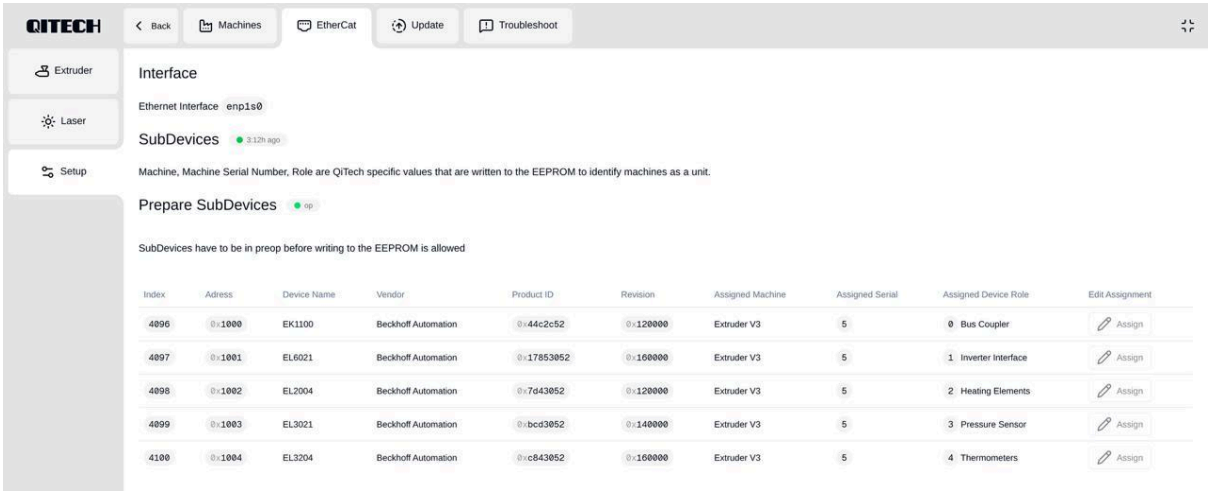
1.3.1 General controls and interface

The QiTech Laser is operated entirely via the QiTech Control software. The machine is accessible on the left-hand sidebar, which makes it possible to control multiple machines simultaneously.

The sidebar on the left switches between the views of the connected machine. *Set Up* provides access to machine identification, EtherCAT device configuration, software updates and system troubleshooting functions.

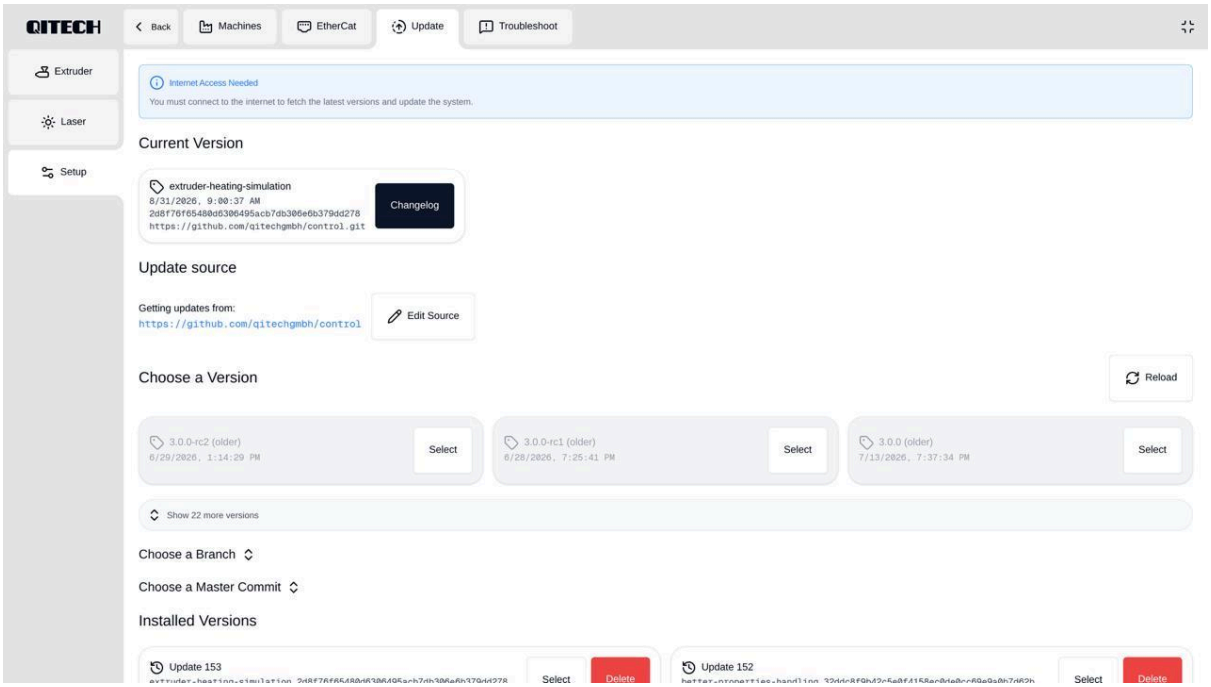


Machines	
Machine	Shows the name and version of the configured machine
Vendor	Shows the manufacturer of the respective machine
Serial	Displays the assigned machine serial number
Error Status	Indicates whether an error is currently detected for the machine



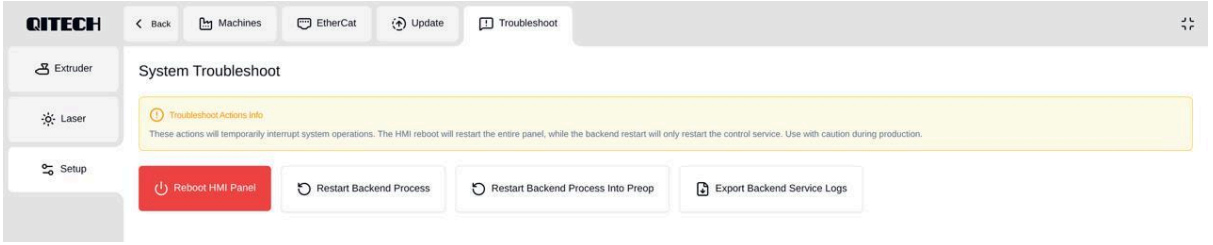
Ethercat	
Ethernet Interface	Shows the Ethernet interface currently used for EtherCAT communication
SubDevices	Displays all detected EtherCAT subdevices connected to the system

Index	Shows the internal index of the respective EtherCAT subdevice
Address	Shows the EtherCAT address assigned to the subdevice
Device Name	Displays the detected EtherCAT device type
Vendor	Shows the manufacturer of the EtherCAT device
Product ID	Displays the EtherCAT product identification number
Revision	Displays the hardware/device revision
Assigned Machine	Shows the machine to which the subdevice is assigned
Assigned Serial	Shows the serial number of the assigned machine
Assigned Device Role	Shows the function of the subdevice within the machine
Edit Assignment	Opens the assignment settings for the selected subdevice



Update	
Current Version	Displays the software version currently installed and running
Changelog	Opens the changelog of the current software version
Update Source	Shows the repository or source from which software updates are retrieved; Edit source if software update source must be changed
Choose a Version	Displays available software versions that can be installed; Select the corresponding software version for installation
Reload	Refreshes the list of available software versions
Choose a Branch	Allows a software branch to be selected manually
Choose a Master Commit	Allows a specific master commit to be selected manually

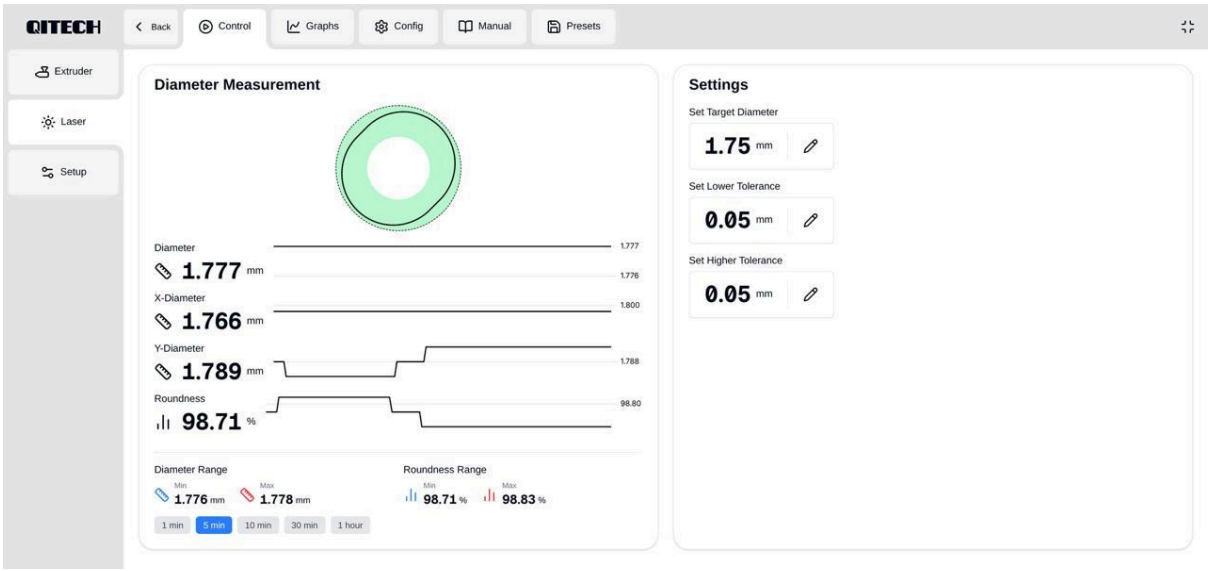
Select Installed Version	Displays software versions already stored on the system; Select or delete a previously installed software version



Troubleshoot	
Reboot HMI Panel	Reboots the complete HMI panel and restarts the user interface
Restart Backend Process	Restarts the backend control process without rebooting the complete HMI panel
Restart Backend Process Into PREOP	Restarts the backend process and places the EtherCAT system into PREOP state
Export Backend Service Logs	Exports backend service log files for troubleshooting and diagnostic purposes

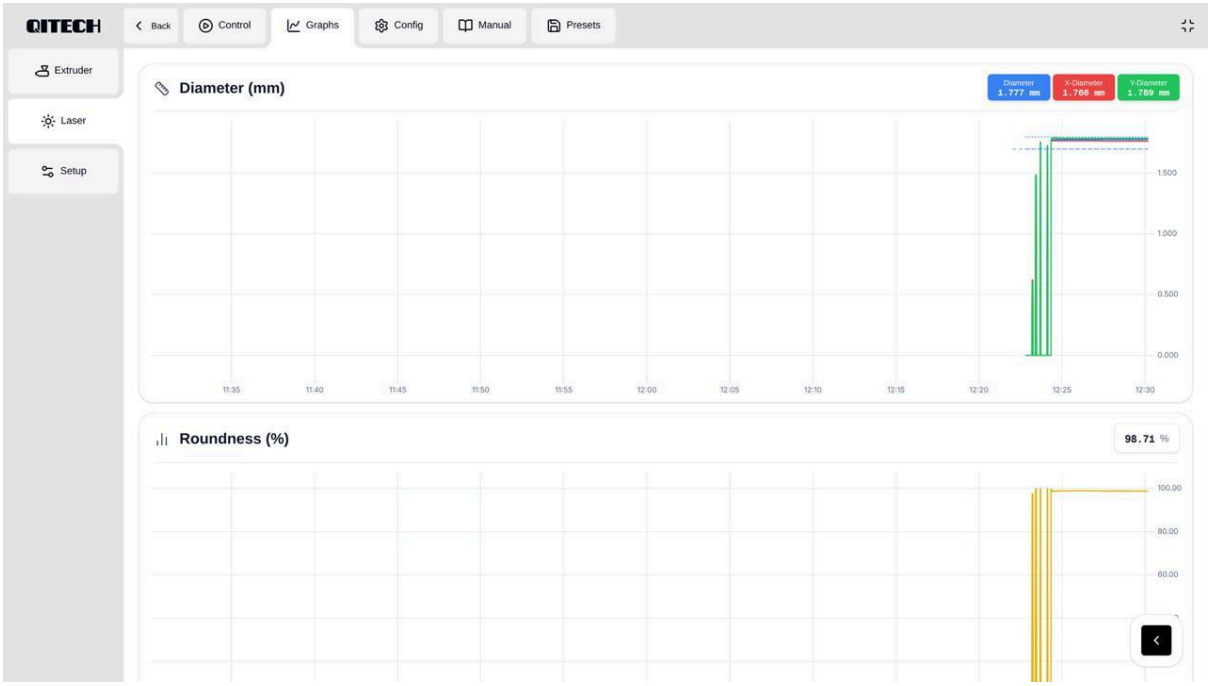
1.3.2 Machine-specific controls and interface

The navigation bar at the top of the screen provides access to the main areas of the QiTech Control software. The *Control* view is the primary operating view, displaying all real-time machine parameters. The *Graphs* view shows process data graphically over time and allows data to be exported in Excel format. The *Config* view provides access to the machine configuration settings. The *Manual* view contains this document, so that all instructions remain available directly at the machine. The *Presets* view is used to save the current parameter set and to load previously stored ones, which allows a proven set of settings to be recalled for a known material. The *Back* button returns to the machine selection screen, and *Fullscreen* expands the software to the full display area.



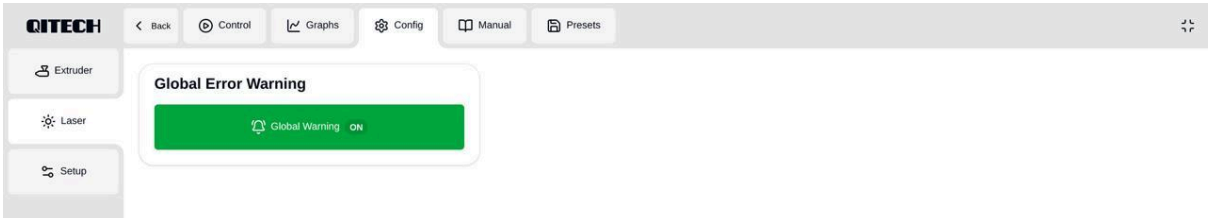
Control	
Diameter Measurement	
Diameter	Displays the current measured filament diameter as the mean value derived from the two measuring axes
X-Diameter	Displays the current filament diameter measured along the X measuring axis
Y-Diameter	Displays the current filament diameter measured along the Y measuring axis
Roundness	Displays the current roundness of the filament cross-section as a percentage; 100 % corresponds to a perfectly circular cross-section
Diameter Range (Min / Max)	Displays the minimum and maximum measured diameter over the selected time range
Roundness Range (Min / Max)	Displays the minimum and maximum measured roundness over the selected time range

Time range (1 min / 5 min / 10 min / 30 min / 1 hour)	Selects the time range over which the measured values and ranges are displayed
Settings	
Set Target Diameter	Sets the nominal target diameter of the filament; adjustable via the edit button
Set Lower Tolerance	Defines the permitted deviation below the target diameter before the measurement is evaluated as out of tolerance; adjustable via the edit button
Set Higher Tolerance	Defines the permitted deviation above the target diameter before the measurement is evaluated as out of tolerance; adjustable via the edit button

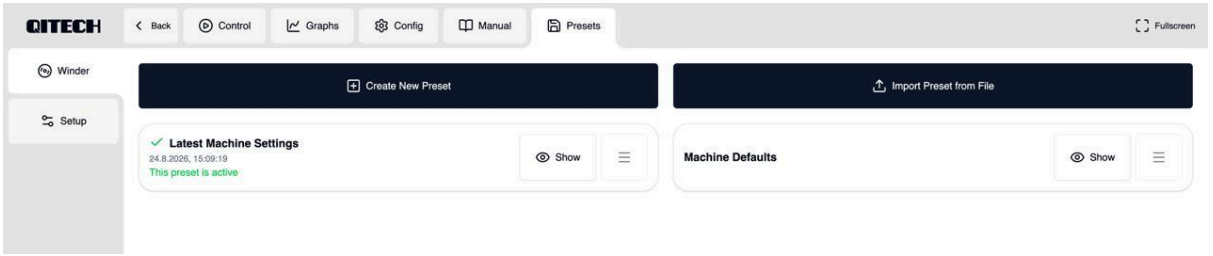


Graph	
Diameter (mm)	Displays the measured Diameter, X-Diameter and Y-Diameter in mm over time
Roundness (%)	Displays the measured roundness of the filament cross-section in percent over time

The graphs can be displayed either as a historical view or as a live view that updates continuously. A time range of up to one hour can be selected. Using the *Add Marker* and *Remove Marker* buttons, specific points in time can be marked manually, named and displayed in the graph. The *Export* button allows the process data to be exported in Excel format.



Config	
Global Error Warning	Enables or disables the global error warning for the machine; when set to ON, detected errors trigger the corresponding warning



Presets	
Create New Preset	Saves the current machine configuration as a new named preset
Import Preset from File	Loads a previously exported preset from a file
Show	Displays the full configuration parameters of the selected preset
Menu (≡)	Opens additional options for the selected preset, such as renaming, exporting or deleting

1.4 Intended use

The QiTech Laser is designed for the non-contact measurement of the diameter of filaments and materials with similar properties, such as cords or wires, with a diameter of 1–3 mm in industrial or commercial indoor environments. It is operated as part of a process line together with the QiTech Extruder, QiTech Airpath or QiTech Aquapath and the QiTech Winder. The machine must be set up on a stable, level surface.

The following conditions of use are prohibited:

- Operation in potentially explosive atmospheres
- Operation in food production environments
- Operation outdoors
- Any use beyond the scope defined in these operating instructions

1.5 Reasonably foreseeable misuse

The following types of use are considered reasonably foreseeable misuse and are prohibited:

Measuring of unsuitable materials:

- Filaments outside the permitted diameter range of 1–3 mm
- Filaments with a surface temperature exceeding 50 °C
- Flat, sharp-edged or pointed filaments

Improper operation:

- Operating the device on an unstable stand or in any position other than the intended one
- Inserting objects other than the intended filament into the measuring gap
- Deliberately looking into the laser beam or exposing the eyes to laser radiation at the measuring gap
- Use of spare parts or components not authorised by the manufacturer
- Operating the device with the housing opened, removed or manipulated
- Operating the device unsupervised or outside of staffed working hours

1.6 Significant modification / other use

Any structural, mechanical or electrical modification to the machine that has not been expressly authorised by QiTech Industries is considered a significant modification. Significant modifications may affect the safety of the machine, render the existing risk assessment invalid and void the CE conformity of the machine. In such cases, a new conformity assessment must be carried out before the machine is put back into service.

Any use of the machine beyond the scope defined in section 1.4 is considered improper use. QiTech Industries and its authorised representatives accept no liability for damage to persons or property resulting from improper use or unauthorised modifications. In such cases, liability rests solely with the operator.

2. Safety

2.1 General safety instructions

These safety instructions apply to all persons working on or with the QiTech Laser. All personnel must be familiar with these instructions and with the information in the following chapters.

Personnel and authorisation

The machine may only be operated by professionally trained and instructed persons. Uninstructed persons and children must not have access to the machine. Repairs and maintenance work may only be carried out by qualified specialist personnel who are familiar with the relevant occupational safety regulations. Work on the electrical equipment of the machine may only be carried out by a qualified electrician.

Personal protective equipment

Appropriate personal protective equipment must be worn at all times. In particular:

- Always wear slip-resistant safety footwear
- Wear close-fitting clothing; use a hairnet if you have long hair
- Do not wear loose clothing, ties, open shirt sleeves or jewellery
- Hearing protection must be worn if the emission sound pressure level at the workplace exceeds 80 dB(A)

Electrical safety

Before carrying out any maintenance, cleaning or repair work, the machine must be switched off and fully de-energised. The following steps must be observed:

- Switch off the device using the on/off switch
- Disconnect the main plug
- Disconnect the LAN connection
- Secure the machine against unintentional restart
- Have the machine de-energised by a qualified electrician
- Earth and short-circuit the machine
- Cover or isolate neighbouring live parts

The mains supply cable may only be connected to a socket outlet with earthing contact or a suitable connection box. Before connection, have the socket outlet or connection box checked by a qualified electrician. Never use the machine with defective, crushed or bare cables.

Laser safety

The QiTech Laser contains a class 2 laser in accordance with DIN EN 60825-1. Visible laser radiation is accessible at the measuring gap during operation.

- Do not stare into the beam
- Do not view the beam directly with optical instruments such as magnifiers or lenses
- The laser warning label must remain present and legible; replace it if it is missing or illegible
- The laser source must not be opened, modified or bypassed

General operational safety

Before starting work, carry out a visual inspection to ensure the machine is in perfect technical condition. The machine may only be used when all protective devices are properly fitted and fully operational. Never leave the machine in operation without supervision.

- Keep the work area clean and free of obstacles; do not place objects on the machine
- Switch off the machine when it is not in use
- Protect the electrical components from moisture
- Check the machine regularly for damage; only use original spare parts authorised by the manufacturer
- Ensure adequate lighting in accordance with applicable national workplace regulations

QiTech Industries accepts no liability for damage caused by failure to observe these operating instructions. The use of non-authorised spare parts or accessories will invalidate the guarantee and warranty.



Danger to life due to technically faulty machine
The machine may only be used when it is in perfect technical condition, in accordance with its intended use and in compliance with these operating instructions. All faults, especially those that could impair safety, must be rectified immediately before starting the machine.

2.2 Pictograms, symbols and warning notices

This chapter provides a comprehensive reference of all safety symbols that operators may encounter in an industrial working environment. Graphical symbols, including warning signs, mandatory signs, prohibition signs, follow ISO 7010. All people working with the QiTech Laser must be familiar with the meanings of these symbols.

The general warning symbol can be used in combination with one of the following signal words:

DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.

NOTICE Indicates important information not related to personal injury. Non-observance may result in damage to property or the environment.

Warning signs		Mandatory signs	
	W001 General warning sign		M001 General warning sign
	W004 Warning of laser beam		M003 Use hearing protection
	W011 Warning of slippery surface		M004 Use eye protection
	W012 Warning of electrical voltage		M008 Use foot protection
	W016 Warning of toxic substances		M009 Use hand protection
	W017 Warning of hot surface		M010 Use protective clothing
	W018 Warning against automatic start-up		M012 Use handrail

	W019 Warning of crushing hazard		M014 Use head protection
	W024 Warning against hand injuries		M021 Switch off before maintenance or repair
	W025 Warning against counter-rotating rollers		
Prohibition signs		**Other pictograms**	
	P001 General prohibition sign		F001 Fire extinguisher
	P002 No smoking		WEEE symbol Separate collection of waste electrical and electronic equipment
	P003 No open flames; fire, open source of ignition and smoking not permitted		
	P022 Eating and drinking not permitted		

	P024 Entering the area not permitted		
	P031 Switching not permitted		
	D-P006 Access not permitted for unauthorised persons		

2.3 Safety devices

The QiTech Laser is equipped with the following safety devices. All safety devices must be fully operational before the machine is started. It is not permitted to operate the machine with any safety device removed, bypassed or disabled. If a safety device is opened or removed for maintenance purposes, it must be correctly refitted and verified before the machine is switched on again.

On/off switch and mains disconnection

The device is switched on and off with the on/off rocker switch on the measuring head. It is not designed as an emergency switching-off device, as the QiTech Laser is a measuring device without hazardous movement. To disconnect the device from the mains supply, switch off the rocker switch and disconnect the mains plug. When disconnected, the device is fully de-energised.

Device housing

The device housing serves as a separating protective device. It restricts access to the laser optics and the electronics, prevents direct contact with live parts, and confines the visible laser radiation to the measuring gap. It may only be opened by qualified personnel after the device has been de-energised.

Acoustic-optical signal device

The measuring head is fitted with an acoustic-optical signal device (buzzer with red indicator light) that acts as a process and tolerance alarm. It is triggered when the measured filament diameter leaves the set tolerance band, defined by Set Lower Tolerance and Set Higher

Tolerance, provided that Global Warning is set to ON in the Config view of the QiTech Control software.

The signal device indicates a deviation of the process from the defined quality limits so that the operator can intervene. It does not stop the process, does not de-energise the device and does not protect against any hazard to persons; it therefore does not replace the operator's supervision of the process. When the signal device is triggered, check the process and, if required, correct the target diameter or the tolerances.

Notice: If safety-relevant changes or defects are detected on the machine, shut it down immediately, secure it against restart and report the issue to the responsible person before resuming operation.

2.4 Qualification of personnel / instruction and training

All persons working with the QiTech Aquapath must have the qualifications appropriate to their assigned tasks. The following personnel categories are defined for the purposes of these operating instructions:

Operator: The operator is the person who uses, adjusts and monitors the machine in day-to-day operation. The operator is characterised by practical experience in handling the machine as well as in-depth knowledge of its functions and the correct course of action in the event of a fault.

Instructed person: An instructed person has been briefed by the operator on the tasks assigned to them and on the potential hazards associated with improper behaviour. The correct execution of all activities carried out by an instructed person is monitored by specialised personnel.

Specialised personnel: Specialised personnel are able to carry out their assigned tasks independently and to recognise and avoid potential hazards, based on their specialist training, knowledge and experience as well as their familiarity with the relevant regulations. It is assumed that specialised personnel have also received the instruction described above.

Qualified electrician: A qualified electrician is a person who, by virtue of their technical training, knowledge and experience as well as their familiarity with the relevant regulations, is able to assess the work assigned to them and recognise and avoid potential hazards related to electrical systems.

Tasks	**Minimum qualification**
Transport	Specialised personnel of the manufacturer
Installation and commissioning	Specialised personnel of the manufacturer

Day-to-day operation	Instructed personnel of the operator
Setting, adjusting and restarting	Specialised personnel of the operator or manufacturer
Troubleshooting and fault rectification	Specialised personnel of the operator or manufacturer
Cleaning and maintenance	Specialised personnel of the operator or manufacturer
Decommissioning	Specialised personnel of the operator

As the operator, you are obliged to ensure that all personnel working on or with the machine are trained and instructed in accordance with these operating instructions before starting work. This includes instruction on all applicable legal and accident prevention regulations as well as on all safety devices on and around the machine. The varying technical qualifications of individual employees must be taken into account when planning instruction sessions.

Instruction must be understood and retained by all personnel. Compliance with instruction content must be verified regularly. Written confirmation of attendance must be obtained from each employee following each instruction session.

Restrictions on employment

The following persons must not work on or with the machine under any circumstances:

- Persons under 18 years of age

2.5 Residual risks

Despite the safety measures integrated into the design and construction of the QiTech Laser, certain residual risks cannot be fully eliminated. The following hazards remain and must be taken into account by all persons working on or with the machine.



Danger! Operation only permitted with all protective devices fitted
The machine must only be operated with all protective devices correctly installed and fully functional. The operator is responsible for ensuring this before each use.



Danger! Laser radiation at the measuring gap
The QiTech Laser contains a class 2 laser. Visible laser radiation is accessible at the measuring gap during operation. Do not stare into the beam and do not direct it towards the eyes. Do not view the beam directly with optical instruments such as magnifiers or lenses.



Danger! Electric shock
Before carrying out any repair, set-up or maintenance work, the machine must be fully de-energised. Do not resume work until the absence of voltage has been confirmed.

3. Transport, installation and commissioning

3.1 Transport

Warning! Risk of injury due to improper transport
When loading and unloading the machine, do not place any limbs under machine parts when setting them down. Do not stand under suspended loads or between moving loads and walls or other machine parts. Wear appropriate personal protective equipment during all transport operations.

Before transport, the following steps must be carried out:

- Switch off the device and disconnect the mains plug
- Disconnect the Z-Link line
- Secure all loose cables and accessories to prevent damage during transport
- If the stand is bolted to the floor, release the floor fixing before moving the device

For transport, suitable lifting equipment must be used. The permissible load-bearing capacity of the lifting equipment must meet the weight of the device. After transport, inspect the device for any damage that may have occurred during transit before putting it into service.

Caution! Risk of damage due to incorrect installation and commissioning
Errors during installation and commissioning can cause considerable damage to property. Installation and commissioning may only be carried out by the manufacturer's employees.

3.2 Requirements for the installation site

Before installing the device, ensure that the installation site meets the following requirements:

- The surface must be level, solid and capable of supporting the weight of the device
- The location must be used exclusively for industrial or commercial purposes
- The location must be as fire-free as possible and equipped with a fire extinguisher
- Sufficient clearance must be maintained around the device to allow safe operation and unobstructed access for maintenance work
- Adequate lighting must be provided in accordance with applicable national workplace regulations
- The ambient conditions at the installation site must lie within the permitted range (see section 1.2 Technical specifications)
- The device must not be installed in the immediate vicinity of equipment causing strong electromagnetic interference, and must be protected from excessive vibration, dust and corrosive atmospheres

3.3 Basic installation

The QiTech Laser is delivered fully assembled on its stand. After positioning the device at the intended installation site, carry out the following steps:

- Position the device at the intended installation site in accordance with the requirements
- Adjust the stand to the required working height and secure it
- If required, bolt the stand to the floor to secure the device against unintentional movement
- Route all cables in a way that they cannot be tripped over
- Ensure that the on/off switch is freely accessible

The device is delivered with the latest available version of the QiTech Control software installed.

3.4 Electrical connection

Danger! Electric shock due to improper electrical connection
All work on the electrical connection must be carried out by a qualified electrician. Ensure the mains socket is de-energised before connecting the machine.

Before connecting the device to the mains supply, carry out the following steps:

- Verify that the mains voltage and frequency at the installation site correspond to the specification of the device
- Ensure that the mains socket is equipped with an earthing contact
- Only use original fuses with the prescribed current ratings to avoid overloading
- Connect the mains plug to the earthing contact socket
- Connect the Z-Link line to the process line to connect to the control display

Ensure that proper earthing of the electrical system is maintained through protective conductor systems at all times.

3.5 Commissioning

Before switching on the device for the first time, carry out a full visual inspection to verify that:

- All protective devices are correctly fitted
- All cables are undamaged and correctly routed
- The on/off switch is in its “OFF” position
- The stand is secured at the working height and, if applicable, bolted to the floor

Once all checks have been completed satisfactorily, the device may be switched on.

4. Operation

4.1 Starting the device

A visual inspection must be carried out every working day before the device is started. Verify that all protective devices are correctly fitted, all cables are undamaged and the on/off switch is in its “OFF” position.

- Connect the mains power supply
- Switch the on/off switch to the “ON” position
- Turn on the process line display
- Select Laser in the left column of the QiTech Control software

4.2 Measuring

In line operation, the filament runs from the QiTech Extruder to the cooling unit and through the measuring gap of the QiTech Laser to the QiTech Winder. The device measures the filament diameter and transmits the values to the QiTech Control software.

- Select Control in the function bar
- Set the nominal diameter using Set Target Diameter
- Set the permitted deviation using Set Lower Tolerance and Set Higher Tolerance
- Guide the filament on the guide rollers through the measuring gap
- The measured Diameter, X-Diameter, Y-Diameter and Roundness are displayed in real time
- If the measured diameter leaves the tolerance band and Global Warning is set to “ON”, the acoustic-optical signal device is triggered

All settings can be adjusted during operation without interrupting the measurement.

4.4 Stopping and switching off

To shut down the device completely:

- Switch the on/off switch to the “OFF” position
- Disconnect the mains plug

Notice: If safety-relevant changes or defects are detected during operation, switch off the device immediately, secure it against restart and report the issue to the responsible person before resuming operation.

5. Maintenance and inspection

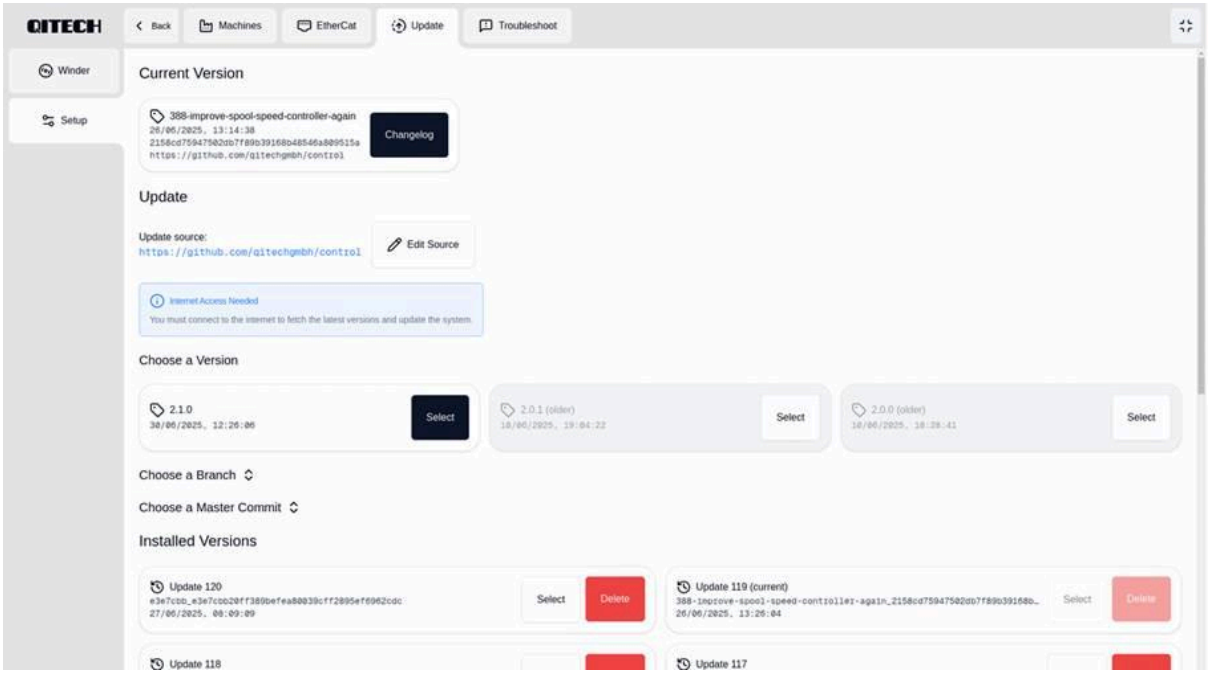
5.1 Updating the QiTech Control software

Software updates may contain bug fixes, new functions or security improvements. It is therefore recommended to check regularly whether updates are available and to install them promptly.

To perform a software update, proceed as follows:

- Ensure the machine has an active internet connection
- Open the QiTech Control software and navigate to *Setup* and *Updates*
- Available updates will be displayed; select the desired update with the *Select* button
- The display will shut down and restart automatically; the updated software version will be installed upon restart

Previous software versions and the complete update history can also be viewed under *Setup* and *Updates*.



Notice: Modifying the software source or installing unofficial software versions may compromise the safety and functionality of the machine and will invalidate the guarantee and warranty. Only official updates from QiTech Automatics GmbH may be installed.

5.2 Mechanical maintenance

Before any cleaning or maintenance work, switch off the device, disconnect the mains plug and disconnect the Z-Link line. Ensure the device is de-energised before starting work.

Cleaning the measuring windows

The measuring windows must be kept clean, as contamination affects measuring accuracy. Check the windows regularly and clean them whenever dust or dirt is visible.

- For light dust, use a suitable lens brush and wipe gently
- For heavier grease or dirt, use a lint-free cloth lightly moistened with a suitable cleaning agent and wipe in one direction to avoid redistributing the contamination
- Do not use abrasive materials or apply excessive pressure to the windows

Retightening screw connections

Check the mechanical fixings of the device and the stand regularly.

- Check that the device is firmly seated on the stand and retighten the mounting screws if necessary
- Check the height adjustment and, if applicable, the floor fixing of the stand for firm seating
- Retighten any loose screw connections to at least hand-tight

Calibration

Calibration ensures the measuring accuracy of the device. It must be carried out whenever the measured value is in doubt. For calibration, the Measured-value display is used, not the control display of the QiTech Software.

- Press "menu" to enter the main menu, select "calibration" and confirm with "OP"
- Enter the diameter of reference rod 1 and confirm with "OP"
- Enter the diameter of reference rod 2 and confirm with "OP"
- Place reference rod 1 in the measuring gap and confirm with "OP"
- Replace it with reference rod 2 and confirm with "OP"
- Select "return" and confirm with "OP" to re-enter the measuring state

6. Troubleshooting

The following table provides an overview of the most common faults, their possible causes and recommended corrective actions. If a fault cannot be resolved using the measures described below, contact QiTech Industries GmbH for technical support.

Before attempting to rectify any fault, ensure that the machine is in a safe condition. For faults requiring access to internal components, switch off the machine and disconnect the main plug.

Fault	Possible cause	Corrective action
Device is not recognised by Control software	Z-Link line not connected	Check and reconnect the Z-Link line
	Outdated software version	Update the QiTech Control software
	Network configuration error	Check network settings
Measured value is implausible or fluctuates strongly	Measuring windows contaminated	Clean the measuring windows (see section 5.2)
	Device not calibrated or calibration in doubt	Recalibrate the device (see section 4.2)
No measured value is displayed	No filament in the measuring gap	Guide the filament through the measuring gap
	Measuring windows heavily contaminated	Clean the measuring windows (see section 5.2)
Acoustic-optical signal device is triggered	Measured diameter outside the tolerance band	Check the process and adjust the target diameter or tolerances if required
	Tolerance set too tight	Adjust Set Lower Tolerance / Set Higher Tolerance in the QiTech Control software
Device does not start	On/off switch in the OFF position	Switch the on/off switch to the ON position

	Mains plug not connected	Check the mains connection and plug in the power supply

7. Decommissioning, disassembly and disposal

When the device has reached the end of its planned service life, QiTech Industries GmbH recommends contacting the manufacturer to arrange for the device to be taken back. If this is not possible, the device must be permanently taken out of service and disposed of in accordance with the instructions in this chapter and in compliance with applicable national regulations.

Disassembly and decommissioning may only be carried out by trained and qualified specialist personnel of the operator.

Decommissioning

Before disassembly begins, the device must be taken out of service by properly switching off, disconnecting the mains plug and disconnecting the Z-Link line, and secured against unintentional restart.

Disassembly

Safety footwear, protective gloves and other protective clothing must be worn during disassembly.

The device consists largely of metal and plastic components, which can be separated by loosening the screw connections. Proceed as follows:

- Remove all cables and set them aside for separate disposal
- Detach the device from the stand by loosening the screw connections
- If the stand is bolted to the floor, release the floor fixing
- Separate the mechanical components from the electronic components
- Remove the laser unit and collect it separately for disposal as an electronic component

Disposal

All components must be separated by material class to enable subsequent recycling or differentiated scrapping. The following materials require separate disposal by an authorised specialist company:

- Electrical and electronic components, including the laser unit

Equipment bearing the WEEE symbol must not be disposed of as unsorted waste. The operator is responsible for ensuring that all waste is separated by material and disposed of via an authorised collection or take-back point in accordance with applicable national regulations.

Notice: Disposal of hazardous materials such as lubricants, oils and electronic components must be carried out in accordance with applicable national environmental regulations. Contact an authorised waste disposal specialist if in doubt.

8. Annexes

Annex A: Rendering of the process line

